GAUNTLET

Don't ask AI what to trade.

Ask it whether your strategy deserves to trade.

Adversarial AI strategy-validation layer for trading strategies. Compiles, backtests, attacks, scores, and gates every strategy before it touches paper money.

ROBUSTNESS SCORE · ILLUSTRATIVE EXAMPLE

PASS · 89.7 / 100

One example strategy's real score. Slide 7 shows an actual verified run, start to finish.

MIDE_XOL SEPTEMBER 3, 2026

THE PROBLEM

MOST AI TRADING TOOLS ANSWER THE WRONG QUESTION.

They tell you what to trade. They don't tell you whether your strategy has earned the right to trade. A strategy that looks great on one backtest can still be fragile.

CURVE-FIT

Looks great on one backtest, breaks on the next.

REGIME-BLIND

Works in one market era, fails in another.

ONE SHOCK AWAY

A single volatility spike from blowing up.

Nobody attacks the strategy before real capital does.

THE PIPELINE

IDEA, STRATEGY, ATTACK, ROBUSTNESS, GATE, PAPER TRADE, MONITOR.

1 › 2 › 3 › 4 › 5 › 6 › 7

1 IDEA
Plain English strategy description

2 STRATEGY
LLM compiles it into strict rules

3 ATTACK
5 adversarial stress tests

4 ROBUSTNESS
Weighted score, 0 to 100

5 GATE
PASS, CONDITIONAL, or BLOCK

6 PAPER TRADE
Real order submitted via Alpaca

7 MONITOR
Watches for the failure it already flagged

Every stage is auditable. Every run, score, gate decision, and order is written to the database.

THE 5 CRASH TESTS

FIVE INDEPENDENT ATTACKS. ONE ROBUSTNESS SCORE.

01 · 25%

Historical Robustness

Does it hold across different market eras, or only on average?

02 · 20%

Parameter Sensitivity

Does a small input tweak collapse performance? Signals curve fitting.

03 · 20%

Drawdown Resilience

How bad is the worst decline, and how long does it stay underwater?

04 · 20%

Volatility Stress

How does it perform specifically in the highest volatility windows?

05 · 15%

Market Reversal

Does exit logic limit damage when the trend suddenly flips?

Weighted 25 / 20 / 20 / 20 / 15 into a single Robustness Score, 0 to 100. No LLM in the math, anywhere.

SAFETY ARCHITECTURE

THE LLM NEVER TOUCHES THE MATH.

RULE 1

Scoring is deterministic.

LLMs compile English into rules, and explain already computed numbers in prose. They cannot alter a score, a verdict, or the gate decision. Every number is plain pandas and numpy math.

RULE 2

Live trading is hard blocked.

Not a config flag. Enforced at three separate points in the code: client initialization, the risk gate, and order submission. No single point of failure could enable it.

Every external dependency, Alpaca and the LLM provider, has exactly one point of contact in the codebase.

TECH STACK

A LEAN STACK, ONE JOB PER FILE.

BACKEND

- Python
- FastAPI
- pandas / numpy
- alpaca-py
- Gemini (google-genai)
- SQLite

FRONTEND

- React 19
- TypeScript
- Vite
- Tailwind v4
- Recharts

Deployed on Railway or Render for the backend, and Vercel or Netlify for the frontend.

LIVE, NOT MOCKED

VERIFIED END TO END THIS SESSION.

Real Alpaca paper trading data. Real Gemini API calls. Every line below is an actual response, not a mockup.

OK	compile-strategy	NVDA · SMA 10/30 · 5% size · benchmark QQQ
OK	crash-test	robustness_score 61.5 · failed: parameter_sensitivity
OK	explain	LLM-generated plain-English analysis of the failure
OK	risk-gate	CONDITIONAL · 61.5 is between 60 and 80: reduced size, monitored
OK	paper-trade	BUY NVDA · $2,500 notional, halved for CONDITIONAL
OK	monitor	no alert · recent drawdown -7.49%

Nothing in the UI is mocked. Every number came from a live call to the backend.

SCOPE & ROADMAP

NARROW ON PURPOSE. HONEST ABOUT WHAT'S NEXT.

NON-GOALS

- No live-money trading.
- No options.
- No ML price prediction.
- MVP strategy scope: SMA crossover only.

NEXT UP

- More strategy types.
- Run-history view in the dashboard.
- Automated test suite.

Live-money trading is hard-blocked at three separate points in the backend, in every build of this project.

GAUNTLET

Don't ask AI what to trade.

Ask it whether your strategy deserves to trade.

github.com/midexol/Gauntlet

Thank you. Questions?